Then $\theta = \tan^{-1}\left[\left|\frac{2\sqrt{h^2-ab}}{a+b}\right|\right]$

$= \tan^{-1}\left[\frac{2\sqrt{\frac{25}{4}-6}}{5}\right]$

$\therefore \theta = \tan^{-1}\left(\frac{1}{5}\right)$

Example 3.12

The slope of one of the straight lines $ax^2+2hxy+by^2=0$ is twice that of the other, show that $8h^2=9ab$.

Solution

Let m_1 and m_2 be the slopes of the pair of straight lines

$ax^2+2hxy+by^2 = 0$

$\therefore m_1+m_2 = -\frac{2h}{b}$ and $m_1m_2 = \frac{a}{b}$

It is given that one slope is twice the other, so let $m_2 = 2m_1$

$\therefore m_1+2m_1 = -\frac{2h}{b}$ and $m_1 \cdot 2m_1 = \frac{a}{b}$

$\therefore m_1 = -\frac{2h}{3b}$ and $2m_1^2 = \frac{a}{b}$

$\Rightarrow 2\left(-\frac{2h}{3b}\right)^2 = \frac{a}{b}$

$\Rightarrow \frac{8h^2}{9b^2} = \frac{a}{b}$

$\Rightarrow 8h^2 = 9ab$

Example 3.13

Show that the equation $2x^2+7xy+3y^2+5x+5y+2=0$ represent two straight lines and find their separate equations.

Solution

Compare the equation

$2x^2+7xy+3y^2+5x+5y+2=0$

with

$ax^2+2hxy+by^2+2gx+2fy+c=0$,

we get,

$a=2,\ b=3,\ h=\frac{7}{2}, g=\frac{5}{2},\ f=\frac{5}{2},\ c=2$

Now $\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = \begin{vmatrix} 2 & \frac{7}{2} & \frac{5}{2} \\ \frac{7}{2} & 3 & \frac{5}{2} \\ \frac{5}{2} & \frac{5}{2} & 2 \end{vmatrix}$

$= 2\left(6-\frac{25}{4}\right)-\frac{7}{2}\left(7-\frac{25}{4}\right)\cdot$

$+\frac{5}{2}\left(\frac{35}{4}-\frac{15}{2}\right)$

$= 2\left(-\frac{1}{4}\right)-\frac{7}{2}\cdot\frac{3}{4}+\frac{5}{2}\cdot\frac{5}{4}$

$= -\frac{1}{2}-\frac{21}{8}+\frac{25}{8} = 0$

Hence the given equation represents a pair of straight lines.

Now consider,

$2x^2+7xy+3y^2 = 2x^2+6xy+xy+3y^2$

$= 2x(x+3y)+y(x+3y)$

$= (x+3y)(2x+y)$

Let $2x^2+7xy+3y^2+5x+5y+2 = (x+3y+l)(2x+y+m)$

Comparing the coefficient of x,

$2l+m=5$ (1)

Comparing the coefficient of y,

$l+3m=5$ (2)

Solving (1) and (2), we get $m = 1$ and $l = 2$

$\therefore$ The separate equations are

$x+3y+2=0$ and $2x+y+1=0$.

Example 3.14

Show that the pair of straight lines $4x^2-12xy+9y^2+18x-27y+8=0$ represents a pair of parallel straight lines and find their separate equations.

Solution

The given equation is $4x^2-12xy+9y^2+18x-27y+8=0$

Here $a=4, b=9$ and $h=-6$

$h^2-ab = 36-36=0$

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